MIL-F-45911A
29 June 1973
SUPERSEDING
MIL-F-45911(WC)
31 March 1966

MILITARY SPECIFICATION

FITTINGS, AUTOMOTIVE TUBE, SPHERICAL SLEEVE (COMPRESSION) TYPE, BRASS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for tube fittings of the spherical sleeve compression type (see 6.1 and 6.3.1).

1.2 Various types of fittings used in the spherical sleeve compression type series are shown on Supplement I to MIL-F-45911.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip)

Military

MIL-V-3 - Valves, Fittings and Flanges (Except for Systems Indicated Herein); Packaging of

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage

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(See Supplement I for list of additional applicable MS sheet form standards.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Society of Automotive Engineers (SAE)

SAE J246 - Spherical Sleeve (Compression) Tube Fittings

(Applications for copies should be addressed to the Society of Automotive Engineers (SAE) Standards Committee, 485 Lexington Avenue, New York, New York 10017.)

American Society for Testing and Materials (ASTM) Standards

ASTM - E18 - Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM - B145 - Leaded Red Brass and Leaded Semi-Red Brass Sand Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

American National Standards Institute

ANSI - B46.1 - Surface Texture; Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

American Trucking Association, Inc.

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee Agency

Uniform Freight Classification

(Applications for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Fittings covered by this specification shall be made of material (see 6.3.2) conforming to QQ-B-626, copper alloy number CA360, half hard for brass bar and extruded shapes, or CA377 for brass forgings. Brass cast fittings shall be made of material conforming to ASTM B145, alloy number 5A, or an equivalent material with zinc not to exceed 16%. It is recommended that cast fittings in sizes below 5/8 tube O.D. not be used for critical applications. Sleeves shall be made of material conforming to QQ-B-626, CA360 brass and annealed to a maximum hardness of Rockwell F70.

3.2 Dimensions. Dimensions and tolerances of the fittings shall be in accordance with the applicable documents listed in Supplement I to MIL-F-45911 or, if not specified, in accordance with Society of Automotive Engineers Standard SAE J246.

3.2.1 Size designations. The fitting sizes are designated by the corresponding outside diameter of the tubing for the various sizes of tube ends and by corresponding standard nominal pipe size for pipe thread ends shown on the applicable documents listed in Supplement I to MIL-F-45911. If not covered by the applicable documents, they shall be in accordance with SAE J246.

3.3 Design and construction.

3.3.1 Contour. Details of contour for the fittings specified in the applicable documents listed in Supplement I to MIL-F-45911 shall be optional with the manufacturer, provided the dimensions are in accordance with the applicable documents of Supplement I and the serviceability of the fitting is not impaired. Wrench flats on elbows and tees shall be optional. When fittings are made from bar stock, the maximum body contour may be increased to that of the maximum thread size of the fitting, plus finished stock.

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3.3.2 Passages. Where passages in straight fittings are machined from opposite ends, the offset at the meeting point shall not exceed 0.015 inch. The cross-sectional area at the junction of passage in angle fittings shall not be less than that of the smallest passage. At the manufacturers option, all passages may conform with the smallest diameter specified, or the appropriate end may be counterbored to the larger diameter for depth defined in the "S" dimension of SAE J246.

3.3.3 Wall thickness. Unless otherwise specified, the wall thickness at any point on the fittings shall not be less than the thickness established in SAE J246.

3.4 Threads.

3.4.1 Straight threads. Straight threads shall conform to the Unified Screw Thread Standard, Class 2A and 2B, in accordance with the Screw-Thread Standard for Federal Services, Handbook H28.

3.4.2 Thread eccentricity tolerances. The various thread elements of Class 2A external and Class 2B, modified, internal threads on tube fittings shall be concentric within the limitations specified in SAE J246.

3.4.3 Pipe threads. Unless otherwise specified (see 6.2), tapered pipe threads shall conform to the Dry Seal American Standard Taper Thread (NPTF) as specified in Handbook H28.

3.5 Identification. Unless otherwise specified (see 6.2), all fitting nuts shall be permanently and legibly marked "AIR BRAKE". Size and location of the marking shall be optional with the manufacturer. Other identification shall be as specified (see 6.2).

3.6 Surface roughness. All sealing surfaces for the compression sleeve shall be smooth, except that annular tool marks up to 100 microinches (RHA) maximum will be acceptable. Roughness Height Average (RHA) shall be interpreted in accordance with the American National Standards Institute, ANSI B46.1. Surfaces not usually machined shall have a surface finish conforming to standard commercial practice.

3.7 Pressure. Cast fittings shall withstand a minimum air pressure of 300 pounds per square inch (P.S.I.) for a minimum of 30 seconds without leakage. Unless otherwise specified, extruded and forged fittings shall also withstand 300 P.S.I. pressure for 30 seconds without leakage.

3.8 Workmanship. Workmanship shall conform to the best commercial practice to produce high quality fittings. Fittings shall be free of machining fluids, chips, hanging burrs, loose scale, and slivers which might become dislodged in usage and all other defects which might affect their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms. Inspection terms used in this specification are defined in MIL-STD-109.

4.2 Lot formation. A lot shall consist of all fittings of the same material, type, style, heat treatment, and size, produced by the same manufacturer under essentially the same conditions, and submitted for acceptance at one time.

4.3 Special inspections and tests.

4.3.1 Visual and dimensional examination. A random sample of fittings shall be taken from each lot in accordance with MIL-STD-105, Inspection Level II. The Acceptable Quality Level (AQL) shall be as specified in Table I.

4.3.2 Pressure tests.

4.3.2.1 A random sample of cast fittings shall be taken from each lot for the pressure test of 4.5.2 in accordance with Level S-2 of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.3.2.2 Unless otherwise specified, a random sample of extruded and forged fittings shall be taken from each lot for the pressure test of 4.5.2 in accordance with Level II of MIL-STD-105. The Acceptable Quality Level (AQL) shall be 6.5 percent defective.

4.3.3 Packaging and packing. Sampling for inspection and test of packaging and packing shall be in accordance with MIL-V-3.

4.3.4 Sleeve test. A random sample of sleeves shall be taken from each lot for the sleeve hardness test of 4.5.3. The lot acceptance shall be based on the sampling requirement of MIL-STD-105, at Inspection Level S1. If one sleeve is rejected, the lot shall be rejected.

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4.4 Quality conformance inspection.

4.4.1 Visual and dimensional. Each fitting selected as specified in 4.3.1 shall be examined to verify compliance with this specification. Examination shall be conducted in accordance with Table I.

TABLE I - Classification of Defects

Categories	Defects	Inspection Method
Critical	None defined	
Major	AQL = 1.0 percent defective	
101	Fittings, not type or style specified	Visual
102	Sealing surface, not as specified (3.6)	SIE*
103	Dimensions (3.2 and 3.2.1)	SIE*
104	Threads, not as specified (3.4.1, 3.4.2, 3.4.3)	SIE*
105	Counterbore, not as specified (3.3.2)	SIE*
106	Bores drilled from each end exceed specified tolerance at drill meeting point; size of fluid passage less than specified; fluid passage of angle fittings, not as specified (3.3.2)	SIE*
107	Wall thickness of fittings, less than dimensions and tolerances specified (3.3.3)	SIE*
Minor	AQL = 4.0 percent defective	
201	Marking, not as specified (3.5)	Visual
202	Quality of fittings poor; fittings contain machining fluids, chips, hanging burrs, loose scale and slivers	Visual

* Standard inspection equipment

4.5 Quality conformance tests.

4.5.1 Material. Unless otherwise specified, the contractor shall be required to furnish certification that materials conform to the requirements of 3.1.

4.5.2 Pressure. The fittings, selected in accordance with 4.3.2, shall be assembled with tube(s), sleeve(s) and nut(s) and the tapered threads plugged or capped, as applicable. The tube length shall be at least eight times the tube diameter. The assembly shall be submerged in water and 300 p.s.i. air pressure shall be applied for at least 30

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seconds. Any sign of leakage in any part of the assembly, as indicated by air bubbles, shall be cause for rejection.

4.5.3 Sleeve hardness test. The fittings selected in accordance with 4.3.4 shall be tested for hardness in accordance with ASTM - E18 to determine conformance to 3.1.

4.6 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with Section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Fittings shall be packaged in quantities as specified in accordance with the Level A packaging requirements of MIL-V-3.

5.1.2 Level C. Fittings shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. The fittings shall be packed in accordance with the applicable Level A requirements of MIL-V-3.

5.2.2 Level B. The fittings shall be packed in accordance with the applicable Level B requirements of MIL-V-3.

5.2.3 Level C. The fittings shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. Fittings covered by this specification are intended for use with copper tubing in the piping of air brake systems on vehicles.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. MS Part Number from applicable document listed in Supplement I to MIL-F-45911.
c. Threads, if other than specified (3.4.3).
d. Selection of applicable level of packaging and packing required (5.1 and 5.2).
e. Additional or deleted marking, if required (3.5).

6.3 Interchangeability.

6.3.1 These fittings were formerly known as air brake tube fittings of the compression sleeve type. To assure satisfactory performance, tapered sleeve compression type fitting components should not be intermixed with these spherical sleeve compression type fitting components when assembling connections in areas where both types are available.

6.3.2 Material similarity. SAE CA360 and CA377 materials are identical in all respects with QQ-B-626, CA360 and CA377 materials.

6.3.3 Substitutability. Table II compares the nomenclature and style of fittings used in MIL-F-45911(WC) with the Military Standards used in this specification.

TABLE II

MIL-F-45911(WC)		MIL-F-45911A
Nomenclature	Style	Military Standard
Adapter	1	MS39179
Adapter	2	MS39181
Elbow	1	MS39182
Elbow	2	MS39186
Elbow	3	MS39184
Elbow	4	MS39185
Nipple	1	MS39187
Tee	1	MS39188
Tee	2	MS39190
Tee	3	MS39191
Tee	4	(Deleted, no requirement)
Tee	5	MS39193
Tee	6	MS39189
Tee	7	MS39192 (Cancelled)
Sleeve	1	MS39197
Nut	1	MS39196

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Custodians:
Army - WC
Navy - None
Air Force - 82

Review Activities:
Army - ME
Navy - None
Air Force - None
DSA - CS

User Activities:
Army - AT, EL, ME
Navy - AS, MC
Air Force - None

Preparing Activity:
Army - WC

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